

FastStrings V1

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Fixed-Capacity High-Performance Strings for C and C++

Version 1.0 (2026 Edition)

Overview

FastStrings is a deterministic, fixed-capacity string layer for C and C++.

It eliminates repeated ``strlen`` scanning by maintaining explicit length metadata.

It preserves full compatibility with standard C APIs such as ``printf``.

Core Design

```
typedef struct {
```

```
uint32_t max_len;
```

```
uint32_t cur_len;
```

```
} vb_meta_t;
```

The visible string remains a standard ``char[]``.

Declaration

```
#define DCL(name, size) \
```

```
char name[size]; \
```

```
vb_meta_t dv_##name = { (size) - 1, 0 }
```

Key Operations

- ``SET(name, "literal")``

- ``CPY(dest, src)``

- ``CAT(dest, src)``

- ``LEN(name)``

- ``CMP(a, b)``

Performance

FastStrings avoids redundant memory scanning.

Typical tight-loop benchmarks show significant speedup compared to ``strlen + memcpy``.

Reducing unnecessary memory traversal improves CPU efficiency and can reduce energy usage in high-frequency workloads.

Safety

- Fixed capacity
- Automatic truncation
- Always null-terminated
- No heap allocation (C version)

Scope

FastStrings is a memory abstraction only.

File formats are handled separately by the VB Record layer.